dsv4-run-2502.05795 · 52 rounds · 6.2 of 8 H100-h

PLAN & FETCH · r0-11

● [plan] LLaMA-130M: Pre-LN perplexity 26.73 vs Pre-LN+LNS 25.76 r0

○ $ git clone https://github.com/lmsdss/LayerNorm-Scaling repo r5

BUILD · r12-18

○ $ uv pip install … transformers==4.44.2 tokenizers==0.19.1 # aarch64 wheels r17

DEBUG & PATCH · r19-26

○ $ NORM_TYPE=LNS POST_NUM=3 ../.venv/bin/python -c … LlamaForCausalLM(cfg) r19

● UnboundLocalError: cannot access local variable 'present_key_value' r19

○ $ diff /repro/reference/supplement/… peft_pretraining/modeling_llama.py r21

○ 427c427 … elif norm_type == 'cod': … elif norm_type == 'LNS': r21

○ *`elif norm_type == 'LNS'` is DEAD CODE (since 'lns' != 'LNS')* decisive r22

○ $ edit_file … replace_all for `norm_type == 'LNS'` → `norm_type == 'lns'` r25

RUN & TRAIN · r27-45

○ *outputs differ from Pre-LN (max abs diff 1.87) confirming the scaling…* r27

● $ NORM_TYPE=LNS … torchrun_main.py … --num_training_steps 20000 # 3.03 GPU-hours GPU r41

● $ NORM_TYPE=pre … torchrun_main.py … --run_name "130m_res_pre_lr0.001" GPU r44

○ *LNS: PPL = 24.93 (paper: 25.76) … Pre-LN: PPL = 25.07 (paper: 26.73)* r45

VERIFY & REPORT · r46-51

○ $ write_file evidence/REPORT.md r47

lns ppl 24.93 vs paper 25.76; pre-ln 25.07 vs 26.73 **audit 8, reproduced ✓**

● plan ○ *agent* ○ command ● GPU run ● error ○ output ▭ decisive round rN = round